



Molecular dynamics interaction study on natural GLP-1 agonists for diabetes mellitus treatment

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Abstract

Diabetes mellitus is one of the human diseases with the highest morbidity and mortality rate in the world, being of the utmost importance to find alternative treatments to manage the complications of the disorder. GLP-1 is an incretin hormone that enhances insulin secretion and compensates insulin deficiency. In this article, through molecular engineering, we are able to analyse the binding affinity of potential natural GLP-1 agonists by dynamic interactions and their ADMET toxicity profiles of the ligands as potential GLP-1 agonists. In our study, thirty compounds were selected for blind docking simulations, the five best-ranked binding ligands proceeded with subsequent molecular dynamics simulations.

Introduction

Diabetes mellitus is a chronic metabolic disorder characterized by persistent hyperglycemia resulting from insulin resistance or defects in insulin secretion and/or action, and it is associated with high morbidity, mortality, and a substantial global economic burden^{1,2}. GLP-1 is an incretin hormone secreted by enteroendocrine L-cells in the distal small intestine and colon, where it activates GLP-1 receptors. Physiologically, GLP-1 enhances glucose-dependent insulin secretion and suppresses glucagon release, thereby contributing to glycemic control^{3,4}. Therefore, therapeutic targets such as GLP-1⁵ are currently highly sought after as medications for diabetes management. In general, GLP-1RAs are preferred over DPP-4 inhibitors because they provide greater reductions in HbA1c and induce clinically meaningful weight loss, as demonstrated in clinical trials. In contrast, DPP-4 inhibitors produce only modest increases in endogenous GLP-1 levels, resulting in smaller glycemic improvements and minimal effects on body weight⁶. For example, GLP-1 receptor signaling is currently being investigated for its ability to alleviate neuropathic pain mechanisms and to mitigate cardiometabolic risks associated with metabolic dysfunction-associated fatty liver disease (MAFLD)^{4,7}. In this context, GLP-1⁶ have gained increasing attention due to their role in glucose homeostasis and the development of novel pharmacological strategies^{1,7}. In this study, we aim to identify natural compounds that can act as DPP-4 inhibitors using molecular engineering techniques, employing molecular modeling and docking approaches to screen for natural compounds that may interact with DPP-4, predicting their binding affinities and potential effects. This approach aims to discover low-cost, natural alternatives for diabetes treatment, offering new therapeutic options for managing the disease and its complications.

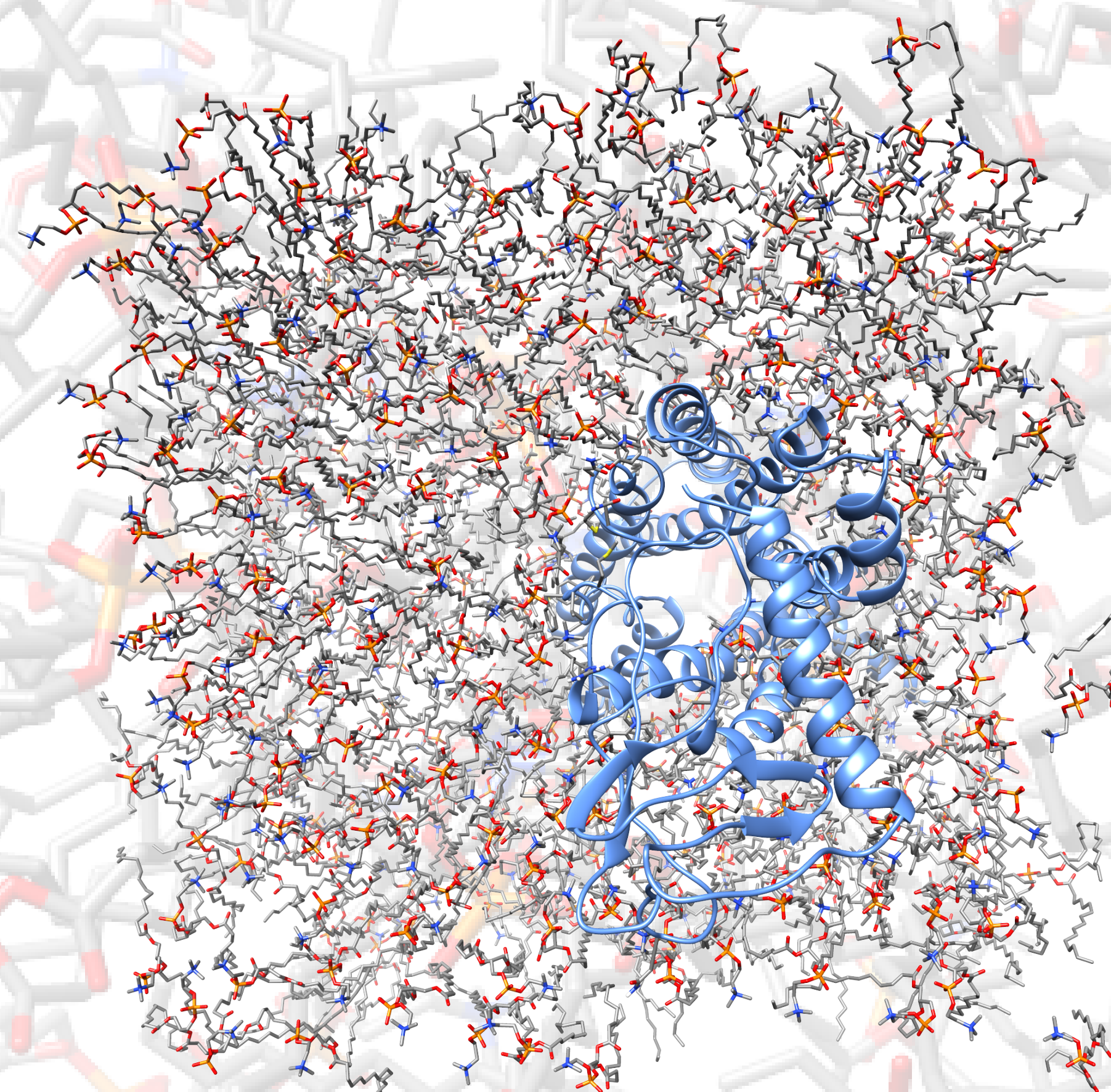


Figure 2: Final GLP-1 structure.

Methodology

The present study focuses on several platforms and software tools, including the Protein Data Bank (PDB), Chimera, GROMACS, Gaussian, Web of Science, LaTeX, JabRef. The GLP-1 protein structures had already been analyzed using ChimeraX and GROMACS software. A search for known natural ligands resulted in a total of 30 GLP-1 agonists candidate compounds. SMILES notation and three-dimensional structures of the ligands were retrieved and verified through PubChem, Lotus, and Coconut databases. The ligands were previously prepared through structural cleaning and geometric optimization using the Gaussian 16 software. The optimized ligands were converted to the PDBQT format for molecular docking. During this step, polar hydrogens were added and nonpolar hydrogens were merged according to AutoDock Vina requirements. Docking simulations were then performed using AutoDock Vina, generating 30 binding poses per ligand with a maximum of 1000 steps per run. The 5 best-ranked binding ligands were selected based on binding affinity scores (verbascoside, salvianolic acid B, hamamelitannin, yohimbine and sennoside A) with values between -9.552 to -8.971, which shows they had a good binding potential.

These selected complexes were further processed in Chimera, where hydrogens were added to the ligands to ensure proper protonation. Ligand topologies and parameters were then generated using the LigParGen server, which provided OPLS-compatible force field parameters, including ligand topology files (.itp), and structure files (.gro), for subsequent molecular dynamics simulations. Hirshfeld atomic charges were calculated for the ligands using Gaussian 16. The resulting charges were adjusted to ensure overall charge neutrality of each ligand and were subsequently incorporated into the ligand topology files for molecular dynamics simulations. After that, molecular dynamics simulations were carried out using GROMACS. The systems were subjected to energy minimization, followed by equilibration under NVT and NPT ensembles, and then to steered molecular dynamics (SMD) simulations. RMSD and RMSF analyses were performed on the GROMACS trajectories to evaluate the global stability and residue-wise flexibility of the protein-ligand complexes. Binding free energy calculations were then conducted using the Molecular Mechanics Poisson-Boltzmann Surface Area (MM/PBSA) method based on simulation snapshots to obtain refined binding affinity estimates beyond static docking performed in gmx. MMPBSA.

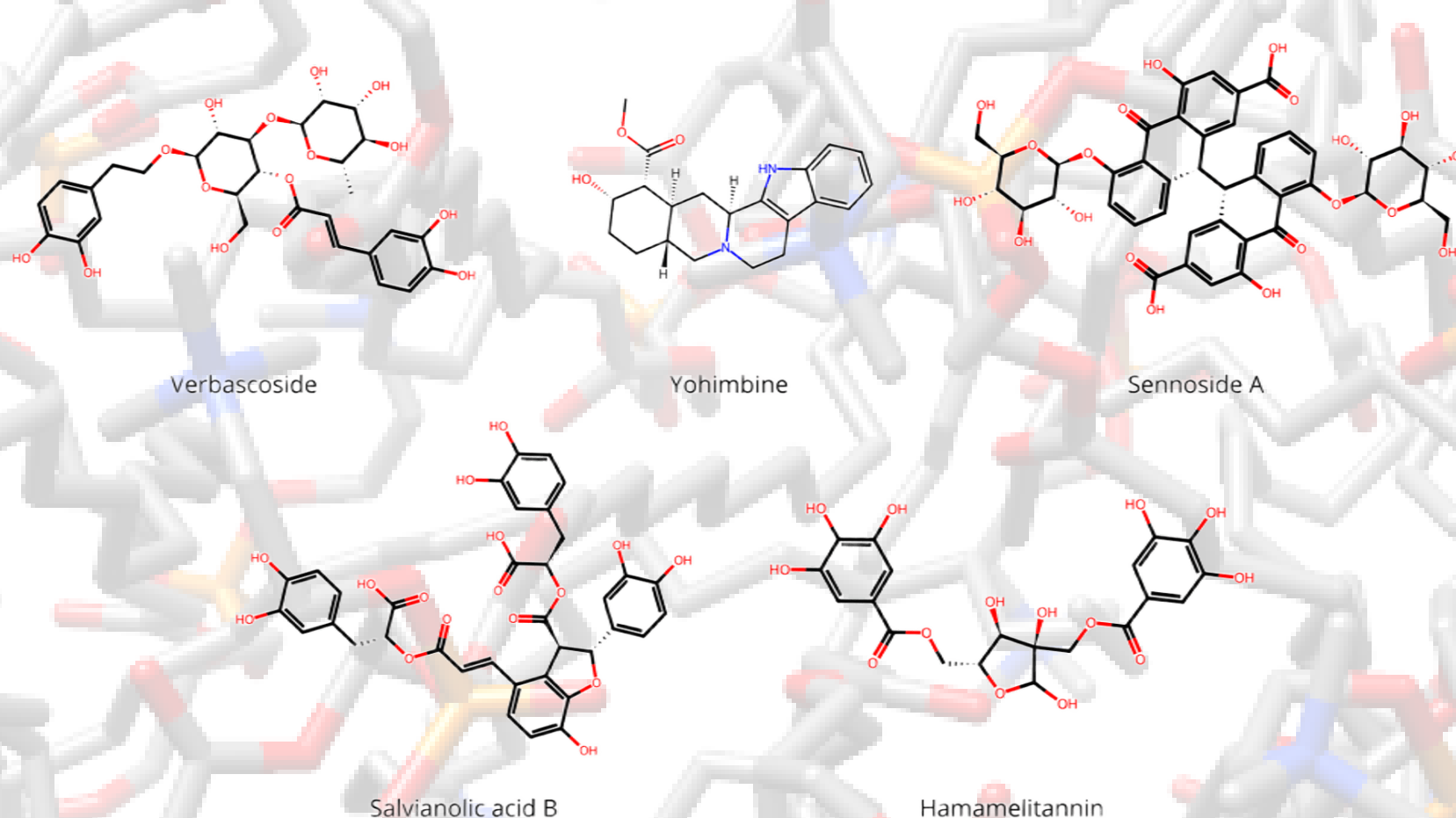


Figure 3: Natural ligands selected for the dynamics.

Figure 2 shows the initial and final stable state of the GLP-1 and Figure 6 shows the RMSD and RMSF of this equilibrium showing an excellent stable conformation.

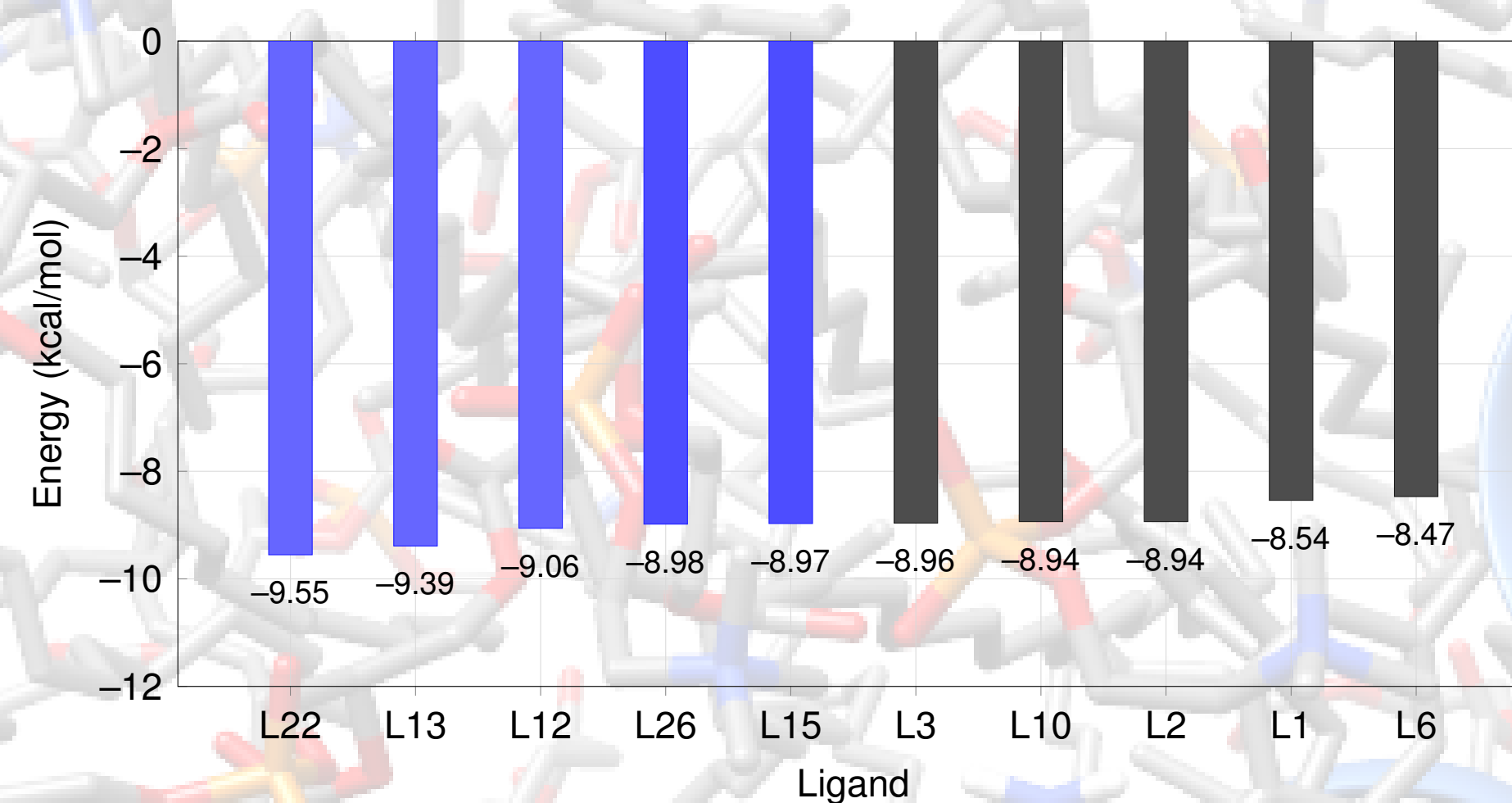


Figure 4: The 10 best results after blind docking.

During the RMSD and RMSF analysis, shown in Figure 6 it was observed that L28 destabilized the protein the most between ns and ns.

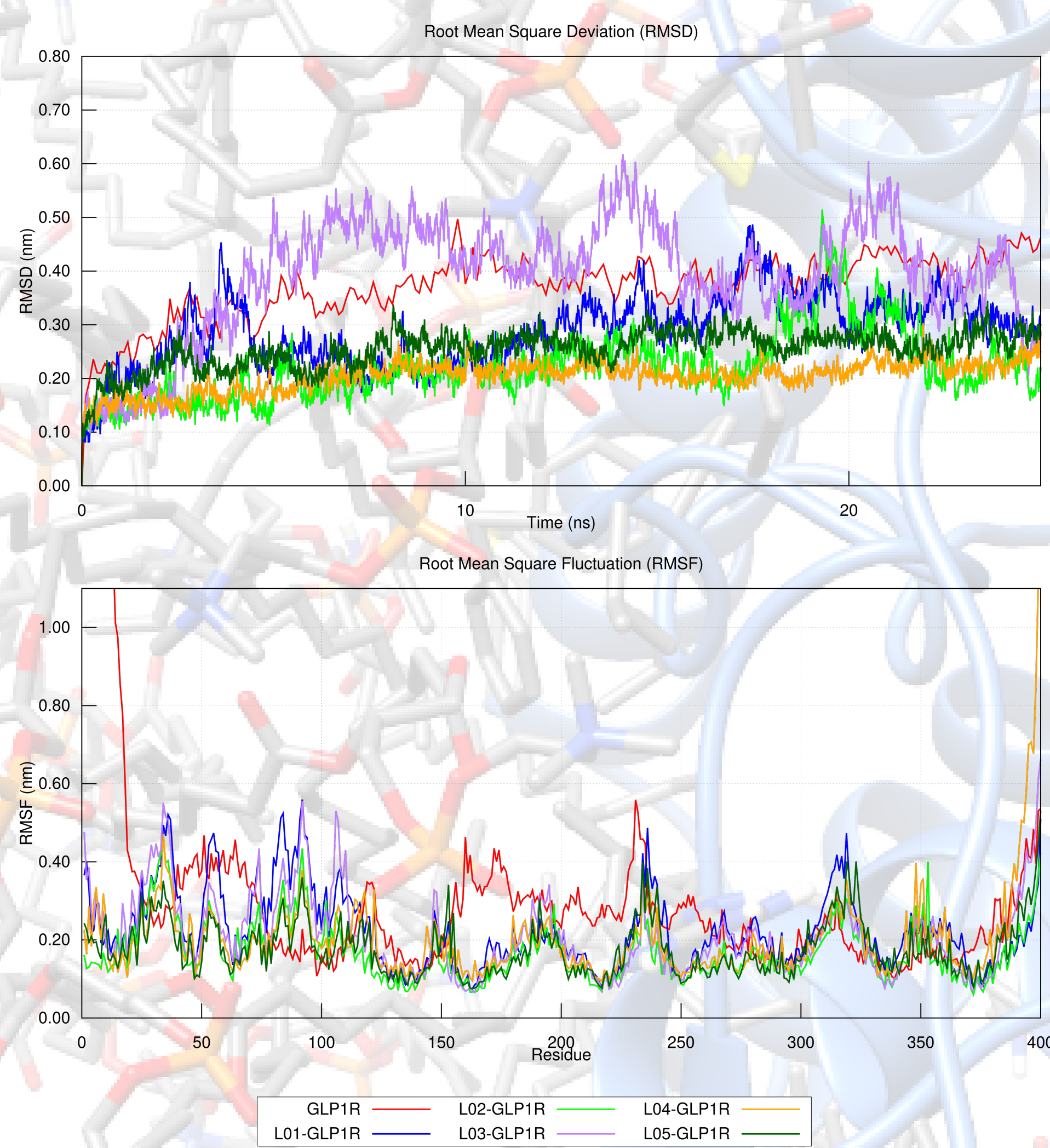


Figure 6: GLP-1-ligands RMSD (top) and RMSF (bottom) after 25 ns.

The reactivity of each ligand was analyzed through RMSD and RMSF and as shown in Figure 6. L3 displayed the most negative energy out of all of them (-42.03), therefore it is the most reactive and stable ligand with GLP-1. L5 had the second most negative binding energy (-16.57), showing a good reactivity with GLP-1. L15 had the third most negative binding energy (-27.51), being also a good candidate for further studies. L22 had the fourth most negative binding energy (-18.67). At last, the L13 had a positive binding energy (416.57), making it the least stable ligand with GLP-1.

Compound	ΔG_{bind}
verbasoside (L3)	-42.03
salvianolic acid B (L13)	416.57
hamamelitannin (L15)	-27.51
yohimbine (L22)	-18.67
sennoside A (L5)	-35.78

The ADMET 3.0 predictions reveal that Sennoside A and Salvianolic Acid B possess prohibitive toxicity profiles marked by predicted liver, kidney, and genetic damage. Hamamelitannin poses specific cardiotoxic (hERG) and cytotoxic risks. Yohimbine appears to be the least systemically toxic regarding organ failure, though respiratory risks are pronounced. Extensive structural modification would be required to mitigate the severe safety liabilities predicted for the majority of these compounds. The toxicity profiles of the top five ligands were evaluated using in silico ADMET 3.0 analysis. The ADMET 3.0 analysis revealed severe safety limitations for the evaluated compounds, with heterogeneity in their toxicity profiles. Sennoside A and Salvianolic Acid B exhibited the most critical toxicity burdens, demonstrating near-maximum probabilities (>99%) for Drug-Induced Liver Injury (DILI), nephrotoxicity, ototoxicity, and genotoxicity. Consequently, despite their therapeutic potential, the pronounced multi-organ toxicity predicted for the majority of these compounds necessitates extensive structural optimization.

Conclusions

This study evaluated the interaction of natural ligands with GLP-1, a key target in diabetes treatment. Following blind molecular docking, 5 potential candidates were chosen based on their binding energies and after molecular dynamics simulations it was revealed that Verbasoside (L03) is the most reactive ligand. Salvianolic acid B (L13) displayed the second best reactivity with GLP-1. However,

Hamamelitannin (L15) had the third reactivity out of all 5 ligands. Yohimbine (L22) was the fourth more negative binding compound, with the best ADMET toxicity profile, being the most safe candidate for further studies. Sennoside A (L26) showed the least negative binding energy and the worst toxicity profile, making it the least favorable candidate for further development. In conclusion, this study highlights the importance of integrating dynamic studies and toxicity profiling in selecting potential GLP-1 agonists for diabetes treatment. Further experimental validation will be essential to confirm these findings.

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